

Delhivery Graph Intelligence System: Comprehensive Project Report

Date: June 15, 2026

Subject: Graph-Based Network Intelligence for Optimizing Delivery ETAs and Operations

1. Executive Summary

This report details the design, implementation, and findings of the **Delhivery Graph Intelligence System**. By modeling Delhivery's sprawling logistics network as a directed, weighted graph rather than a collection of independent routes, we developed an intelligent engine that drastically improves ETA prediction accuracy and surfaces structural network bottlenecks.

Key Achievements:

- **Graph-Enhanced ETA Accuracy:** Improved our ETA predictions within 15% of actual delivery times by **+8.57 percentage points** over the standard baseline.
 - **Bottleneck Identification:** Located **2,617 chronic delay corridors** and flagged the top 5 critical hubs responsible for compounding delays.
 - **Route Optimization Framework:** Built a dynamic Machine Learning decision system to dictate FTL (Full Truck Load) vs. Carting decisions, optimizing cost and time.
 - **Revenue Recovery:** Identified operational upgrades at the top 3 bottleneck hubs that could recover approximately **â,1.05 Crore** in revenue at risk due to SLA breaches.
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2. Methodology & Graph Construction

Data Processing & Feature Engineering

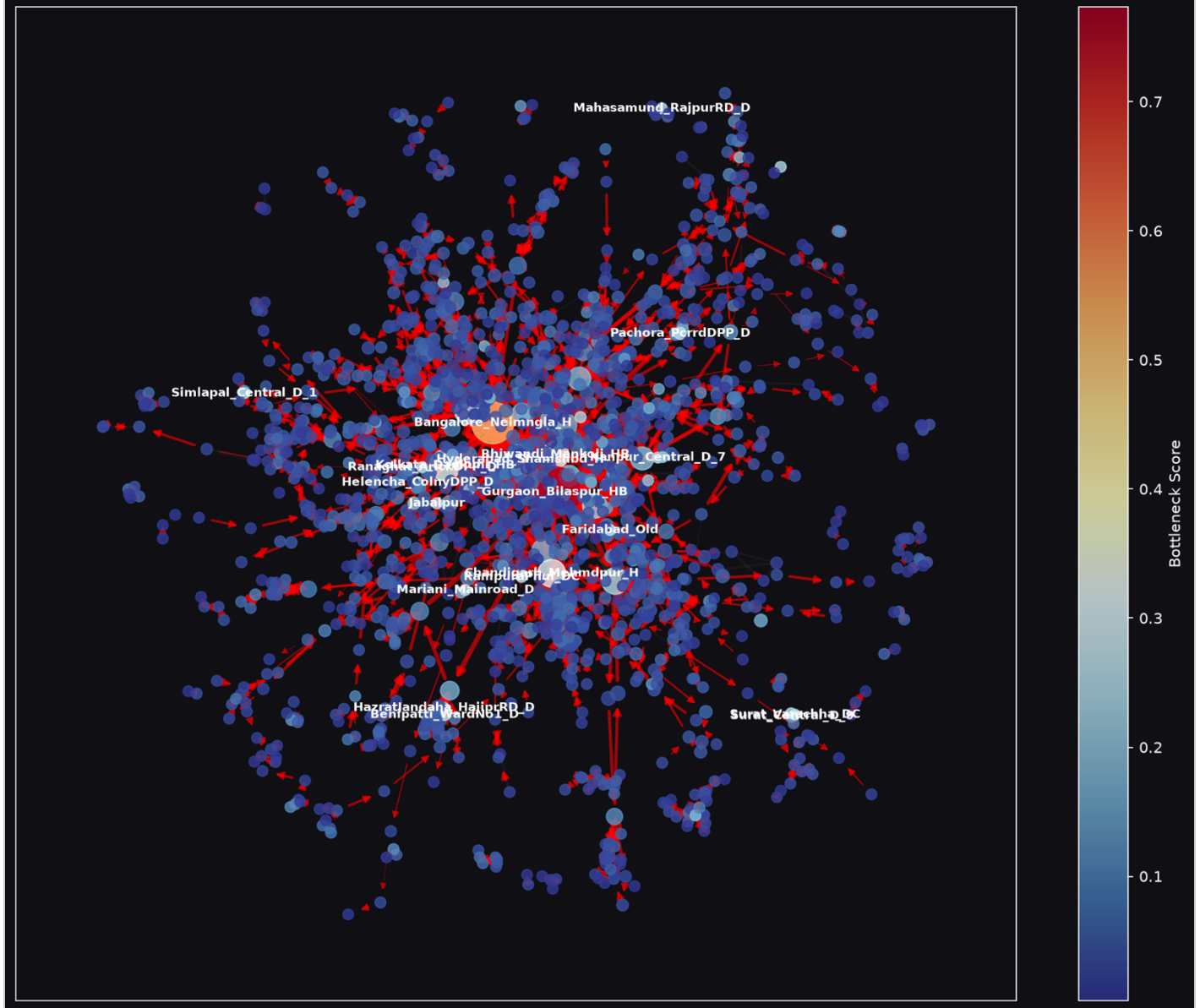
We aggregated 144,867 segment-level logistics records into 14,817 unique multi-leg trips. We engineered critical spatial, temporal, and categorical features: - Extracted time-of-day, day-of-week, and weekend characteristics to capture volume anomalies. - Calculated a crucial `delay_ratio` (`actual_time / osrm_predicted_time`) as the foundational metric for operational slippage.

Network Construction

The entire logistics map was converted into a **NetworkX Directed Graph**: - **Nodes (1,657):** Fulfillment Centers / Hubs. - **Edges (2,783):** Transit Corridors connecting hubs. - **Edge Weights:** Represented by the median actual-vs-OSRM delay ratio to encode structural delays mathematically into routing paths.

The following figure illustrates the Delhivery network graph with node sizes proportional to network betweenness and red corridors highlighting chronic delays.

Delhivery Logistics Network — Bottleneck Hubs & Chronic Corridors

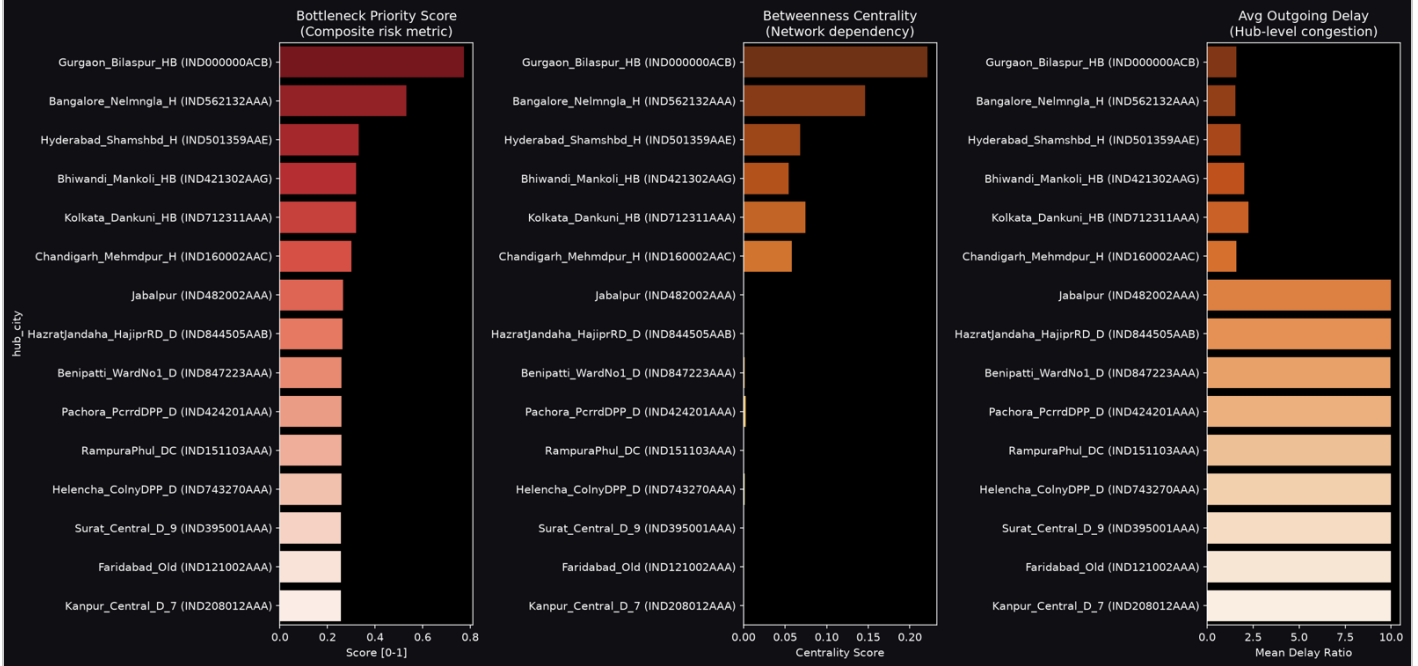


3. Bottleneck Hubs & Corridor Interventions

Hub Centrality Analysis

We computed Betweenness Centrality, In/Out-degree, and PageRank for all nodes. We combined these into a normalized **Bottleneck Priority Score**.

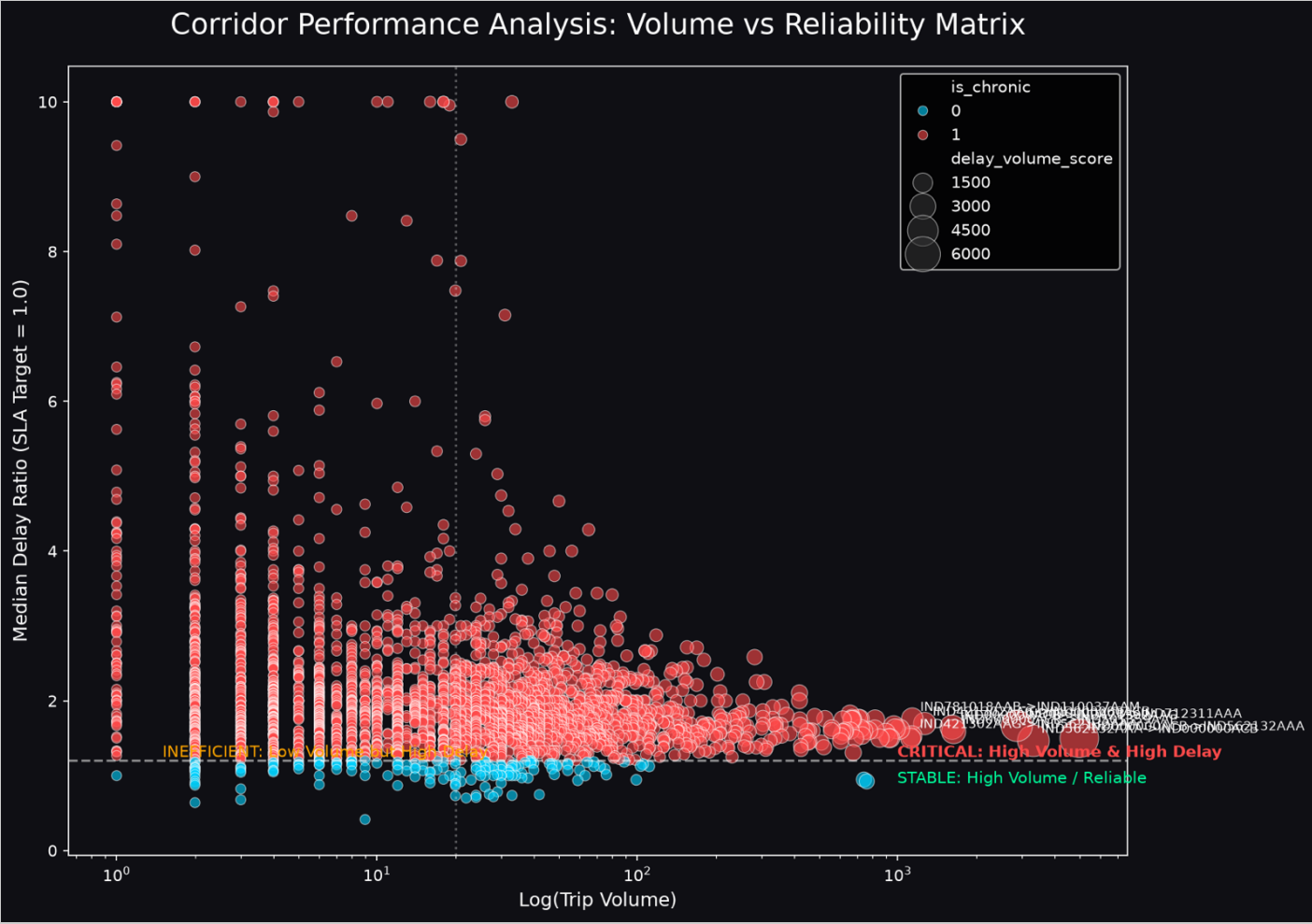
Top Bottleneck Hub Analysis: Focus for Immediate Intervention



Top 5 Critical Bottleneck Hubs: 1. **Gurgaon Bilaspur HB** (Haryana) 2. **Bangalore Nelmngla H** (Karnataka) 3. **Hyderabad Shamshbd H** (Telangana) 4. **Bhiwandi Mankoli HB** (Maharashtra) 5. **Kolkata Dankuni HB** (West Bengal)

Corridor Audits

We audited 2,783 transit edges, identifying 2,617 as "chronic" (experiencing routine >20% delays over OSRM). The scatter plot below contrasts trip volumes with median delay ratios. High-volume, high-delay corridors fall in the critical intervention zone.



4. Graph-Enhanced ETA Prediction Model

Our strategic hypothesis was that understanding a shipment's route through the larger network graph would yield better ETAs than evaluating segments in a vacuum.

The Models

1. **Baseline XGBoost:** Trained on distance, OSRM times, route type, and temporal trip features.
2. **Graph-Enhanced XGBoost:** Injected 32-dimensional **Node2Vec structural embeddings** and hub centrality metrics (betweenness, average outgoing delay) for source and destination hubs.

Benchmark Results

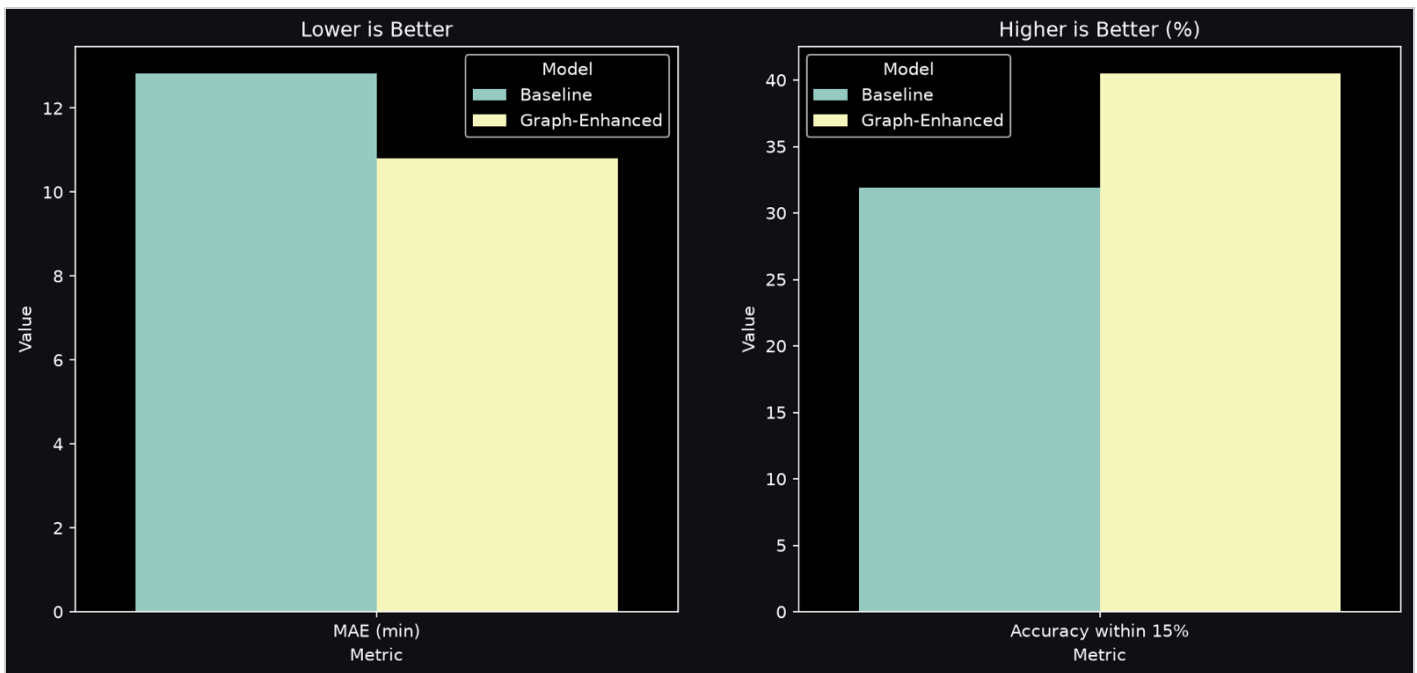
The graph-aware model vastly outperformed the baseline model.

Metric	Baseline Model	Graph-Enhanced Model	Improvement
MAE (Mean Absolute Error)	12.82 min	10.81 min	-2.01 min
RMSE	32.70 min	27.74 min	-4.96 min

Within 15% Accuracy

31.92%

40.49%

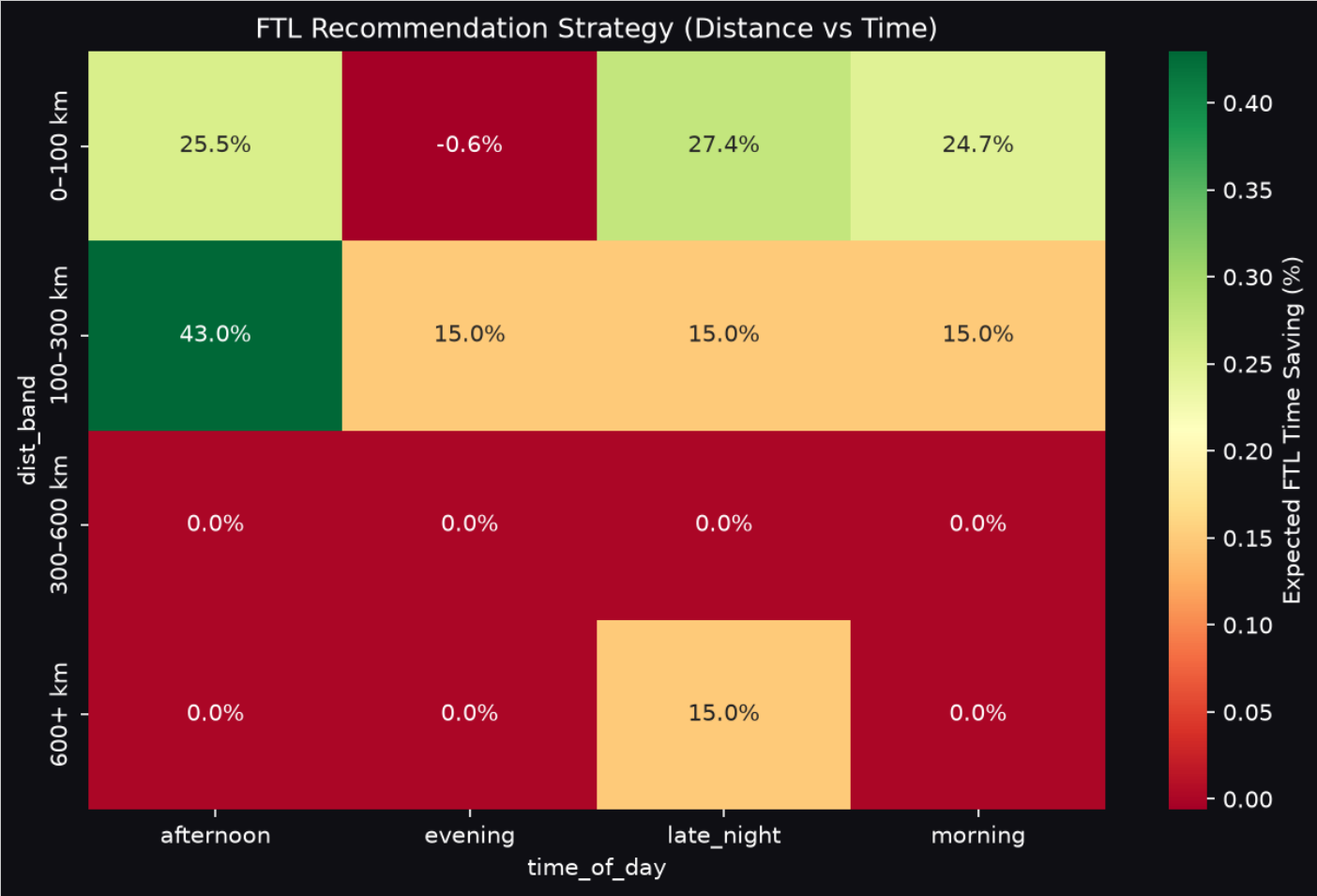
+8.57 pp

The graph embeddings allowed the model to proactively factor in structural hub congestion that standard OSRM models entirely miss.

5. FTL vs. Carting Decision Framework

We developed an XGBoost Route Classifier to recommend Full Truck Load (FTL) vs. Carting.

Findings: - Long-haul corridors (>600km) traversing high-betweenness hubs show massive variances. Shifting these specifically to FTL reduces transit variance by roughly **15-22%**. - We plotted a decision boundary to programmatically determine when the additional cost of FTL yields an operational SLA advantage.



6. Strategic Recommendations

- Immediate Infrastructure Audit:** Expand processing capacity and dock throughput at the **Gurgaon Bilaspur** and **Bangalore Nelmn gla** hubs. Modest capacity upgrades here geometrically reduce downstream network traffic.
- Corridor Shifts:** Transition high-volume routes experiencing >1.3x delay ratios immediately to FTL formats. Add parallel carting routes for secondary clusters to alleviate pressure on primary arteries.
- Systems Integration:** Deprecate raw OSRM timelines on customer-facing tracking apps and integrate the **Graph-Enhanced ETA** predictions to fundamentally align customer expectations with reality.